PROJECT REPORT

TITLE : Employees Turnover Analytics

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PLATFORM : SIMPLILEARN

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1.Abstract

Employee turnover is a critical challenge for organizations as it affects productivity, increases recruitment costs, and impacts overall business performance. The objective of this project is to develop a machine learning-based employee turnover prediction system for Portobello Tech. The project involves data quality assessment, exploratory data analysis, employee clustering, class imbalance handling using SMOTE, and predictive model development using various machine learning algorithms. K-Means clustering was used to segment employees based on satisfaction and evaluation scores, while classification models such as Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting were trained and evaluated using 5-fold cross-validation. The best-performing model was selected based on evaluation metrics such as Recall and ROC-AUC score. Finally, employees were categorized into different risk zones according to their turnover probability, and suitable retention strategies were suggested. The developed solution enables the HR department to proactively identify employees at risk of leaving and implement targeted retention measures.

2.Objectives

The primary objectives of this project are:

1. To perform data quality checks and identify missing values or inconsistencies in the employee dataset.
2. To conduct Exploratory Data Analysis (EDA) and identify factors that contribute significantly to employee turnover.
3. To perform employee segmentation using K-Means clustering based on employee satisfaction and evaluation scores.
4. To address class imbalance in the target variable using the SMOTE (Synthetic Minority Over-sampling Technique) method.
5. To train and evaluate multiple machine learning models using 5-fold cross-validation.
6. To compare model performance using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC.
7. To identify the best-performing model for employee turnover prediction.
8. To predict employee turnover probabilities and classify employees into different risk zones.
9. To recommend appropriate retention strategies for employees based on their predicted turnover risk levels.

3.Tools and Techniques Used

3.1.Programming Language

- Python

3.2.Development Environment

- Jupyter Notebook

3.3Libraries Used

Data Manipulation

- Pandas
- NumPy

Data Visualization

- Matplotlib
- Seaborn

Machine Learning

- Scikit-Learn

Imbalanced Data Handling

- Imbalanced-Learn (SMOTE)

3.4.Techniques Used

Data Quality Assessment

- Missing Value Analysis
- Data Type Verification

Exploratory Data Analysis (EDA)

- Histograms
- Bar Plots
- Correlation Heatmap
- Distribution Analysis

Clustering

- K-Means Clustering
- Elbow Method
- Silhouette Score

Data Preprocessing

- One-Hot Encoding using get_dummies()
- Train-Test Split

Class Imbalance Handling

- SMOTE (Synthetic Minority Over-sampling Technique)

Classification Models

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

Model Validation

- 5-Fold Cross Validation

Model Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve
- AUC Score

Business Analysis

- Employee Risk Categorization
- Retention Strategy Recommendation

4. Dataset Description

The dataset contains information about employees and their workplace behavior.

Features Used

- satisfaction level
- last evaluation

- number project
- average_montly_hours
- time_spend_company
- Work accident
- promotion_last_5years
- sales
- salary
- left (Target Variable)

Target Variable

- left = 1 → Employee Left the Company
- left = 0 → Employee Stayed in the Company

5.Data Quality Check

Data quality check were performed to ensure the reliability of the dataset.

- Checking for missing values df.isnull().sum().
- Data inspection df.info().
- Identified categorical and numerical datatypes.
- Finding the no.of columns and row in the dataset df.shape.
- Performing describe method.

5.1.Observation

No significant missing values were found in the dataset. Therefore, no imputation was required.

6. Exploratory Data Analysis (EDA)

EDA was performed to understand the employee behaviour and factors affecting employee turnover.

6.1.Satisfaction Level Distribution

The distribution plot showed varying levels of employee satisfaction. Employees with lower satisfaction levels were more likely to leave the organization. Most of satisfaction levels were in between 0.4 to 1.0.

6.2.Last Evaluation Distribution

Most employees received moderate to high evaluation scores ranging between 0.5 and 1.0.

6.3.Employee Average Monthly Hours

Most of the employees average monthly working hours are in between 125 to 275 hours.

6.4.Employees Stayed vs Left by Number of Projects

Plotting the distribution with the variables Employee Stay and Number of Projects .Employees who gets less projects and more projects are willing to leave company.

6.5.Correlation Analysis

A correlation heatmap was generated to identify relationships among variables.

6.6.Key insights

- Employees with lower satisfaction levels are more likely to leave.
- Employees handling more projects tend to work more hours.
- Employees with higher performance evaluations often receive more projects.
- Employees who work longer hours generally receive better evaluations.
- Employees who experienced work accidents are slightly less likely to leave.
- Employees who stay longer are only slightly more likely to leave.

6.7.Recommendations

1.Improve employee satisfaction through

- Better life-work balance.
- Career growth opportunities.

2.Monitor workload to prevent burnout.

3.High performers are trusted with additional responsibilities.

4. Managers may associate higher effort with better performance.

5. Work accident employees should receive additional support, benefits, or compensation.

7.Employee Clustering

Importing the KMeans cluster algorithm from sklearn.cluster. KMeans clustering was applied on Satisfaction level, evaluation level and left features in the dataset.

7.1.Clustering process

- Used K-Means algorithm.
- Using 3 clusters as per the project requirement.
- Performing silhouette score to know the quality of clustering.(note: by plotting wcss I got 4 clusters are giving better performance than 3 clusters, silhouette score also little higher than 3 clusters silhouette score)

7.2.Model Silhouette Score

Model silhouette score is 0.43 ,which is not good model or not bad model.

7.3.Cluster Summary

Cluster 0:

- Moderate satisfaction
- Moderate evaluation scores

Cluster 1:

- Lower satisfaction
- Moderate evaluation scores

Cluster 2:

- High satisfaction
- High evaluation scores

7.4.Interpretation

Employees in Cluster 2 appear highly engaged and satisfied, whereas employees in Clusters 0 and 1 may require additional monitoring and support to improve retention.

8. Data Preprocessing and Class Imbalance Handling

The target variable (left) exhibited class imbalance.

8.1.Steps performed

Categorical Encoding

- Sales
- Salary

Were converted into numerical values using get_dummies method.

8.2.Train Test Split

The dataset was divided into:

- 80% training data

- 20% testing data

Using stratified sampling with random state 123 as per project requirement to preserve class distribution.

8.3.SMOTE Application

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data.

- Balance Class distribution.
- Improves model learning.
- Reduce bias towards majority class.

9. Model Training and Cross Validation

Several machine learning models were trained and evaluated using 5-fold cross validation.

9.1.Models used:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

9.2.Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC curve

Cross-validation helped ensure model stability and reduce overfitting.

`10. Model Evaluation

Confusion matrices were generated for all models.

10.1.Precision vs Recall

For employee attrition prediction, Recall was considered more important.

10.2.Reason

Missing employees who are likely to leave (False Negatives) is more costly than incorrectly identifying employees as at risk (False Positives).

Therefore, Recall was selected as the primary evaluation metric.

10.3.ROC-AUC Analysis

ROC curves were generated for all models.

The ROC curve demonstrated the trade-off between:

- True Positive Rate(TPR)
- False Positive Rate(FPR)

AUC scores indicated the models ability to distinguish between employees who are left and those who are stay.

10.4.Best Model

Random Forest is the best model among all three models .because model accuracy score of training and testing data is very high .

10.5.Model Accuracy score:

- On training data : 1.0
- On testing data :0.98966

10.6.Classification Report:

```
              precision    recall  f1-score   support
           0       0.99      0.99      0.99      2286
           1       0.98      0.98      0.98       714

    accuracy                           0.99      3000
   macro avg       0.99      0.99      0.99      3000
weighted avg       0.99      0.99      0.99      3000
```

- High Recall
- Strong ROC-AUC score
- Excellent classification performance

11.Employee Risk Categorization

Using the best model, employee turnover probabilities were predicted and categorized into risk zones.

11.1Safe Zone (Green)

Probability < 20%

Characteristics:

- Highly satisfied employees
- Low turnover risk

Retention Strategy:

- Employee recognition
- Continuous learning opportunities
- Maintain engagement

11.2. Low-Risk Zone (Yellow)

20% ≤ Probability < 60%

Characteristics:

- Moderate turnover risk

Retention Strategy:

- Career development programs
- Regular feedback sessions
- Employee engagement initiatives

11.3. Medium-Risk Zone (Orange)

60% ≤ Probability < 90%

Characteristics:

- Significant turnover risk

Retention Strategy:

- Workload review
- Promotion opportunities
- Salary and benefits assessment
- Managerial intervention

11.4. High-Risk Zone (Red)

Probability ≥ 90%

Characteristics:

- Very high likelihood of leaving

Retention Strategy:

- Immediate HR intervention
- Personalized retention plans
- Compensation review

- Career progression discussions

12.Conclusion

This project successfully developed machine learning models to predict employee turnover at Portobello Tech. Exploratory data analysis identified employee satisfaction, workload, and tenure as important contributors to turnover. K-Means clustering helped segment employees based on satisfaction and evaluation scores. SMOTE effectively addressed class imbalance, and multiple machine learning models were evaluated using 5-fold cross-validation.

Among the evaluated models, the best-performing model demonstrated strong predictive capability based on Recall and ROC-AUC metrics. The proposed risk categorization framework and retention strategies can assist the HR department in proactively identifying at-risk employees and implementing targeted retention programs, thereby reducing employee turnover and improving organizational stability.